

REPLIT PROMPT — YachtWorth: Charter ROI AI Rate Estimator

CONTEXT

Update the existing Charter ROI calculator — specifically the AI rate estimation feature. When user taps "AI Estimate" button, the app calls Claude API with web search enabled. The agent returns ONE specific number (average market rate) that auto-fills the rate input field. Plus a short explanation shown below the field.

HOW IT WORKS IN THE APP

CHARTER RATE

[AI Estimate ✨] [Enter manually]

← user taps "AI Estimate"

→ loading spinner: "Searching market rates..."

→ result auto-fills field:

CHARTER RATE

€ 4,500 per day ← auto-filled, editable

✨ AI Estimate

Based on 12 comparable listings on YachtCharterFleet, Boatbookings and GetMyBoat. Motor yacht 24m 2019, Mediterranean high season (Jun-Sep).

Range found: €3,800–€5,200/day. Recommended: €4,500/day.

[Use this estimate] [Search again] [Enter manually]

User can accept, redo, or override manually.

API CALL

Use Claude API with web_search tool enabled:

```
const response = await fetch("https://api.anthropic.com/v1/messages", {
  method: "POST",
  headers: { "Content-Type": "application/json" },
  body: JSON.stringify({
    model: "claude-sonnet-4-20250514",
    max_tokens: 1000,
    tools: [{ type: "web_search_20250305", name: "web_search" }],
    system: CHARTER_RATE_SYSTEM_PROMPT, // see below
    messages: [
      {
        role: "user",
        content: buildUserPrompt(yacht, roiParams)
      }
    ]
  })
});
```

SYSTEM PROMPT (CHARTER_RATE_SYSTEM_PROMPT)

You are a professional yacht charter market analyst with deep expertise in Medite

Your task: research and return the current market charter rate for a specific yac

SEARCH STRATEGY:

Search these platforms for comparable listings:

- yachtcharterfleet.com
- boatbookings.com
- getmyboat.com
- burgessyachts.com
- fraseryachts.com
- moranyachts.com
- eyos.com
- click&boat.com (for smaller yachts under 15m)

Search query format: "[yacht type] [length]m charter [region] [season] price

COMPARISON CRITERIA:

Find yachts that match ALL of these within tolerance:

- Yacht type: exact match (motor/sailing/catamaran/superyacht)
- Length: within ± 3 meters of specified length
- Year: within ± 5 years of specified year (newer = higher rate)
- Region: exact match
- Season: match the requested season

RATE CALCULATION:

1. Collect minimum 3 comparable listings (aim for 5-10)
2. Note the rate for each (daily or weekly as requested)
3. Remove outliers ($>2\times$ or $<0.5\times$ the median)
4. Calculate: average of remaining rates
5. Round to nearest €100 for weekly, nearest €50 for daily

SEASON DEFINITIONS:

Mediterranean:

- High season: June 15 - September 15
- Shoulder: May 1 - June 14 and September 16 - October 31
- Low season: November 1 - April 30

Caribbean:

- High season: December 15 - April 30
- Shoulder: November 1 - December 14 and May 1 - June 15
- Low season: June 16 - October 31

Northern Europe:

- High season: July 1 - August 31
- Shoulder: May 15 - June 30 and September 1 - September 30
- Low season: October 1 - May 14

Asia-Pacific:

- High season: November - April (varies by location)
- Low season: May - October

Middle East:

- High season: October - April
- Low season: May - September

RATE TYPES:

- Crewed charter (with captain and crew): standard for 15m+ yachts
- Bareboat (without crew): standard for yachts under 15m, sailing yachts
- If user specifies: use what they specify
- Default: crewed for motor yachts 15m+, bareboat for sailing under 15m

OUTPUT FORMAT - CRITICAL:

You MUST respond ONLY with valid JSON. No text before or after. No markdown. No e

```
{
  "rate": 4500,
  "currency": "EUR",
  "period": "day",
  "season": "high",
  "region": "Mediterranean",
```

```

    "confidence": "high",
    "comparables_found": 8,
    "range_min": 3800,
    "range_max": 5200,
    "charter_type": "crewed",
    "sources": ["yachtcharterfleet.com", "boatbookings.com", "burgessyachts.com"],
    "explanation": "Based on 8 comparable motor yachts 22-26m, built 2016-2022, in",
    "seasonal_rates": {
      "high": 4500,
      "shoulder": 3200,
      "low": 2400
    },
    "weekly_equivalent": 31500
  }
}

```

If you cannot find sufficient data (fewer than 3 comparables), set confidence to
 If the region has no active charter market for this yacht type, set rate to null

USER PROMPT BUILDER

```

function buildUserPrompt(yacht, roiParams) {
  const yachtType = yacht.yacht_type || roiParams.yachtType;
  const length = yacht.length_m || roiParams.lengthM;
  const year = yacht.year_built || roiParams.yearBuilt;
  const region = roiParams.region;
  const season = roiParams.season; // 'high' | 'shoulder' | 'low'
  const period = roiParams.ratePeriod; // 'day' | 'week'
  const charterType = roiParams.charterType || 'crewed'; // 'crewed' | 'bareboat'

  return `Find the current market charter rate for this yacht:

```

```

TYPE: ${yachtType}
LENGTH: ${length} meters
YEAR BUILT: ${year}
REGION: ${region}
SEASON: ${season} season
RATE PERIOD: per ${period}
CHARTER TYPE: ${charterType}

```

Search for comparable yachts currently listed for charter in \${region}.
 Return the average market rate as a single number in EUR per \${period} for \${seas
 Also provide the full seasonal rate breakdown (high/shoulder/low).

Respond ONLY with the JSON format specified. No other text.`;

```
}
```

RESPONSE HANDLER

```
async function getAICharterRate(yacht, roiParams) {
  try {
    const response = await callClaudeAPI(yacht, roiParams);

    // Extract text from response content blocks
    const fullText = response.content
      .filter(block => block.type === 'text')
      .map(block => block.text)
      .join('');

    // Parse JSON - strip any accidental markdown
    const clean = fullText
      .replace(/```json/g, '')
      .replace(/```/g, '')
      .trim();

    const data = JSON.parse(clean);

    // Validate required fields
    if (!data.rate || !data.period || !data.seasonal_rates) {
      throw new Error('Invalid response format');
    }

    return {
      success: true,
      rate: data.rate,
      currency: data.currency || 'EUR',
      period: data.period,
      confidence: data.confidence,
      comparables: data.comparables_found,
      rangeMin: data.range_min,
      rangeMax: data.range_max,
      sources: data.sources,
      explanation: data.explanation,
      seasonalRates: data.seasonal_rates,
      weeklyEquivalent: data.weekly_equivalent,
      charterType: data.charter_type,
    };
  } catch (error) {
```

```
    return {
      success: false,
      error: 'Could not retrieve market rates. Please enter manually.',
    };
  }
}
```

UI AFTER RESPONSE

Success — high confidence:

🌟 AI Market Estimate	
RECOMMENDED RATE	
€ 4,500 / day	← large, gold
Range found: €3,800 - €5,200	
Based on 8 comparable listings	
Motor yacht 22-26m · Mediterranean · High season	
SEASONAL RATES:	
High season:	€ 4,500 / day
Shoulder:	€ 3,200 / day
Low season:	€ 2,400 / day
Sources: yachtcharterfleet.com, boatbookings.com, burgessyachts.com	
[✓ Use €4,500/day]	[🔄 Search again]

Success — low confidence (fewer than 3 comparables):

⚠ Limited data	
ESTIMATED RATE	
€ 6,800 / day	
(estimate based on 2 listings)	

Limited listings found for this yacht type and region. Verify before using.
[Use anyway] [Enter manually]

Error / no data:

 Could not retrieve rates
No comparable listings found for this yacht type in the selected region. Please enter the rate manually.
[Enter manually]

AUTO-FILL BEHAVIOUR

When user taps [✓ Use €4,500/day]:

1. Close the estimate panel
2. Auto-fill the charter rate field with the recommended rate
3. Show small badge next to field: ✨ AI estimate · Edit if needed
4. If user had previously entered seasonal rates toggle → also pre-fill shoulder and low season rates from seasonal_rates object
5. Field remains fully editable — user can change the number

LOADING STATE

While API call is in progress (typically 5-15 seconds with web search):

[Searching market data...]

Checking YachtCharterFleet... ✓

Checking Boatbookings... ✓

```
Checking Burgess Yachts...    ⌵
Analysing 8 listings...      ⌵
```

Simulate progress steps with 2-3 second intervals between each line appearing. This reassures user that search is happening.

WHEN TO CALL

Call the AI estimator when:

- User taps "AI Estimate" button
- User taps "Search again" button

Do NOT call automatically — always require user tap.

Cache result for current session: if user navigates away and returns, show cached result rather than searching again.

REGIONS MAPPING

```
const REGION_LABELS = {
  mediterranean: 'Mediterranean',
  northern_europe: 'Northern Europe',
  caribbean: 'Caribbean',
  asia_pacific: 'Asia-Pacific',
  middle_east: 'Middle East',
  global: 'Global / Other',
};

// For 'global' region: search Mediterranean as reference market
// Note in explanation: "Using Mediterranean as reference market for global estim
```

FILE STRUCTURE

```
/utils/
  charterRateAI.js      ← buildUserPrompt(), getAICharterRate(), SYSTEM_PROMPT

/components/CharterROI/
```


IMPORTANT NOTES

1. The system prompt instructs Claude to return ONLY JSON — parse carefully
 2. web_search tool must be enabled in the API call — without it Claude has no current data
 3. Cache the result in component state — don't re-call for the same parameters
 4. The recommended rate is the AVERAGE of comparables, rounded to nearest €50 (daily) or €500 (weekly)
 5. If user changes yacht parameters after getting estimate → clear cached result and show "AI Estimate" button again
 6. Timeout: if API call takes more than 30 seconds → show error and fallback to manual entry
 7. Rate is always in EUR regardless of region — conversion note not needed for yacht charter market
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YachtWorth Charter ROI — AI Rate Estimator Uses Claude API with web_search tool Returns single recommended rate + seasonal breakdown